Bayesian Temporal Tensor Factorization-Based Interpolation for Time-Series Remote Sensing Data With Large-Area Missing Observations

Haixu He, Jining Yan, Lizhe Wang, *Fellow, IEEE*, Dong Liang, Jianyi Peng, and Chengjun Li

***Abstract*— Land surface temperature (LST) is widely used in the field of time-series remote sensing. However, due to the influence of cloud cover, the large area of LST data observation is missing, which seriously affects the later data analysis. In the past research, various effective interpolation methods have been developed, but they usually cannot effectively interpolate the image data with large observation missing. In this article, a new method for interpolating these missing data called Hilbert tensor rearrangement with Bayesian temporal tensor factorization (HTR-BTTF) is proposed. This method requires tensor rearrangement of remote sensing data, combined with BTTF method for interpolation. In order to evaluate the performance of our method, we select three real study areas with different climates, Wuhan, Harbin, and Kunming LST data during day and night, and add cloud covers with different sizes to the cold and warm season layers each year. BTTF, inverse distance weighted (IDW), harmonic analysis of time series (HANTS), and GapFill are used as comparison methods. Root-mean-square error (RMSE) is a comprehensive evaluation index of interpolation results. Experiments have shown that HTR-BTTF is an effective method for interpolating missing observations, which is better than other methods. In the simulation experiment of the largest cloud cover size, on average, the RMSE of the data filled using the HTR-BTTF method was 17.2% lower than that of BTTF and 54.9% lower than that of the GapFill method, and it shows good robustness and high accuracy.**

***Index Terms*— Bayesian factorization, Hilbert curve, interpolation, land surface temperature (LST), time series.**

I. INTRODUCTION

LAND surface temperature (LST) refers to the temperature of the ground. After the heat energy of the sun is radiated and reaches the ground, part of it is reflected and part of it is absorbed by the ground and heats it. The temperature obtained by measuring the temperature of the ground is the surface temperature. Surface temperature is a key factor in surface physical processes at regional and global scales. It is also an important parameter for studying material exchange and energy exchange between the surface and the atmosphere. In studying land use change analysis [1], [2], estimating air temperature and soil moisture [3], [4], and detecting urban heat island effects [5] and other areas, it has a wide range of applications. The National Aeronautics and Space Administration (NASA) provides global Moderate-Resolution Imaging Spectroradiometer (MODIS) data with high spatial and temporal resolution [6] and these data have been widely used worldwide. However, when cloud pollution factors cause low-quality or missing data on land satellites, which is a very common situation, especially for cloudy areas, it is actually impossible to accurately reconstruct continuous time-series datasets.

In the face of large areas of data missing in MODIS images, many algorithms have been developed to solve them. They can be divided into the following categories [7]. The first category includes geostatistical spatial–spectral interpolation methods, such as kriging interpolation and its variants. Its principle is to use the autocorrelation relationships existing within the observation data to express the nonobservation point value with the weighting function of the observation point value. This method was originally used to estimate the grade of ore bodies. In recent years, this method has been widely used in the fields of hydrogeology and petroleum geology. Kilibarda *et al.* [8] used spatiotemporal regression kriging to interpolate the average temperature, maximum temperature, and minimum temperature in space and time. Shiode and Shiode [9] proposed the interpretation method of extended kriging and inverse distance weighting (IDW) to calculate more accurate surface temperatures. In a multispectral image, the spectral information of different bands can be used to reconstruct the missing data in a specific band [10]. Wang *et al.* [11] proposed using the analytical relationship between Terra MODIS band 6 and 7 to restore Aqua MODIS band 6. This method is usually used to fill in the missing information caused by the sensor. When the spatial–spectral interpolation method is applied to the area of high cloud cover, the interpolated image is prone to distortion, and it cannot solve the situation when a single image is completely missing. Moreover, the error in the central area of the cloud cover is generally higher than that in the edge area.

The second category is temporal-based methods that use the information obtained by missing pixels at different times to fill in the gap. Zhang *et al.* [12] adopted the linear time method.

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For each missing pixel, the change rate of LST between the nearest missing time point and the subsequent eight-day data period was selected to construct a linear equation to solve. Ma *et al.* [13] proposed the linear memory vector recurrent neural network (LIME-RNNs) model that is based on the residual network to propose residuals and vectors to method the missing and preexisting item associations, to realize the automatic capture of the time-series dynamic dependence, and to improve the filling effect. Luo *et al.* [14] used an improved generative adversarial network to achieve multivariate time-series interpolation, which improved the interpolation accuracy. The harmonic analysis of time-series (HANTS) method is used to smooth the normalized difference vegetation index (NDVI) and the interpolation of invalid data [15], [16]. Zhou *et al.* [17] used HANTS to reconstruct the NDVI time series and other surface variables to prove the effectiveness of the HANTS algorithm. However, the temporal-based methods lack spatial reference information, and the global consistency of the interpolation results is not good.

The third category is methods based on spatiotemporal methods. This approach makes good use of the correlation between the temporal dimension and the spatial dimension. Gerber *et al.* [18] proposed a GapFill method based on specific image sorting and quantile regression, which users can access and use for free [19], [20]. Malambo and Heatwole [21] proposed a contour-based interpolator that uses a spatial window to recover lost information. Li *et al.* [22] used spectrum and time information simultaneously to propose a method based on sparse representation to reconstruct the analysis method of missing data in remote sensing images. The time cost of pixel-by-pixel interpolation based on spatiotemporal information is generally high, especially when interpolating missing data in a large area, but this type of method can refer to more information in multiple dimensions, and the effect is usually the best.

The aforementioned problems motivate our present research. In this study, we add a tensor rearrangement module to the Bayesian temporal tensor factorization (BTTF) method, and a method with high accuracy and good robustness is obtained, which is suitable for interpolation of missing observations in a large area. In the space dimension, the Hilbert filling curve is used to disperse the missing observations in a large area. In the temporal dimension, the time-series rearrangement method is used to extract the periodic characteristics of the time series. Finally, the BTTF method is used for interpolation and we call it Hilbert tensor rearrangement with BTTF (HTR-BTTF). The novelty of our work is given as follows.

1) We fuse spatiotemporal information, use the Hilbert space-filling curve to solve the large-area lack of spatial reference information, and use the temporal rearrangement method to make full use of periodic features.
2) We propose to adopt the Bayesian tensor factorization (BTF) method to interpolate time-series remote sensing data with missing observations. This method can fit the distribution of the data for interpolation without setting many prior parameters.
3) We propose to adopt early stopping to reasonably reduce the number of method iterations, and the restack tensor method is used for interpolation to reduce method running time.

The detailed method flow will be introduced in Sections III and IV.

This article is organized as follows. Section II introduces the BTTF method. Section III introduces the HTR-BTTF method in detail. Section IV introduces the experimental data and experimental design, as well as the results obtained in the simulation study. Section V discusses the advantages and improvements of the HTR-BTTF method. Section VI ends with a conclusion.

II. Preliminaries

BTF is a commonly used interpolation algorithm, which is used in the recommendation system [23], [24], spatiotemporal traffic data interpolation [25], [26], and image restoration [27]–[29]. In the application of spatiotemporal data, Chen *et al.* [30], [31] proposed a method of applying BTTF of the vector autoregressive (VAR) process to the time factor matrix to interpolate and predict the spatiotemporal traffic data and achieved good results.

A. Method Introduction

The time-series remote sensing data can be defined as a tensor $D \in \mathbb{R}^{t\times m\times n}$, where m and n represent the spatial side length of the study area, respectively, and t represents the time-series length. We use $(i, j, t) \in \Omega$ to index the observed entries in tensor D. Then, randomly initialize three low-rank factor matrices, U, V, and X, without missing data according to the CANDECOMP/PARAFAC (CP) decomposition process of tensors [32]. We know that the first dimension of D shows the time-series characteristics, and the second dimension and the third dimension of D show spatial characteristics, so we define X as temporal factor matrix and U and V as space factor matrices. We assume that each missing item in the tensor obeys an independent Gaussian distribution, and we assume that each observed entry d in D follows a Gaussian distribution with precision τ [33], [34]

$$d_{i,j,t} \sim \mathcal{N}\left(\sum_{r=1}^{R} u_{ir} v_{jr} x_{tr}, \tau^{-1}\right). \tag{1}$$

Under the Gaussian assumption, τ represents the degree of noise in the observed data. Here, it will be assumed that the standard deviation τ obeys the elastic conjugate Gamma distribution to improve the robustness of the method

$$\tau \sim \text{Gamma}(\alpha, \beta). \tag{2}$$

In order to method the tensor data correctly, we define the prior distributions on the spatial factor matrices U and V. Specifically, suppose that the prior distributions of row vectors u_i and v_j in the U and V factor matrices are multivariate Gaussian distributions

$$\begin{aligned} u_i &\sim \mathcal{N}\left(\mu_u, \Lambda_u^{-1}\right) \\ v_j &\sim \mathcal{N}\left(\mu_v, \Lambda_v^{-1}\right). \end{aligned} \tag{3}$$

In a Bayesian setting, it is assumed that the hyperparameters obey the conjugate Gaussian–Wishart distribution. This will enhance the robustness of the method and speed up the convergence speed when performing method inference using sampling algorithms [30]. The prior distributions of μ and Λ are defined as follows:

$$\begin{aligned}(\mu_u, \Lambda_u) &\sim \text{Gaussian} - \text{Wishart}(\mu_0, \beta_0, W_0, \upsilon_0)\\ (\mu_\upsilon, \Lambda_\upsilon) &\sim \text{Gaussian} - \text{Wishart}(\mu_0, \beta_0, W_0, \upsilon_0)\end{aligned} \tag{4}$$

where υ_0 represents the degrees of freedom, W_0 represents the ratio matrix of $R \times R$, and $\mu_0 \in \mathbb{R}^R$ is a mean vector that can be defined as a zero vector.

The processing of the temporal factor matrix is different from the space factor matrix. The time factor matrix exhibits time-series characteristics, so the VAR can be used to predict the missing data in the time series. The autoregressive method assumes that there is a linear dependence between variables. For the tth moment of multivariate time-series data $X \in \mathbb{R}^{R\times T}$, there is the following linear expression:

$$x_t = \sum_{k=1}^{c} A_k x_{t-k} + \epsilon_t \tag{5}$$

where A_k is an $R \times R$ coefficient matrix and ϵ_t is a Gaussian noise vector.

We can introduce matrix A and vector υ_t

$$A = [A_1, \ldots, A_c]^T \in \mathbb{R}^{N\times N} \tag{6}$$

$$\upsilon_t = \begin{bmatrix} x_{t-1} \\ \vdots \\ y_{t-c}\end{bmatrix} \in \mathbb{R}^{(Nc)}. \tag{7}$$

The VAR method can be rewritten as $x_t = A^T \upsilon_t + \epsilon_t$. Also, the lag set is defined as $\mathcal{L} = \{h_1, \ldots, h_k, \ldots, h_c\}$. Thus, the temporal factor matrix X can be written as

$$x_t \sim \begin{cases}\mathcal{N}(0, I_R), & \text{if } t \in \{1, 2, \ldots, h_c\}\\ \mathcal{N}(A^T \upsilon_t, \Sigma), & \text{otherwise.}\end{cases} \tag{8}$$

We place the conjugate matrix normal inverse Wishart (MNIW) prior on the coefficient matrix A and the covariance matrix Σ

$$\begin{aligned}A &\sim \mathcal{MN}_{(Rc)\times R}(M_0, \Psi_0, \Sigma)\\ \Sigma &\sim \mathcal{IW}(S_0, \upsilon_0).\end{aligned} \tag{9}$$

Parameter τ assumes the gamma distribution

$$\text{Gamma}(\hat{\alpha}, \hat{\beta}). \tag{10}$$

It is basically impossible to directly solve the posterior distribution, so Salakhutdinov and Mnih [24] proposed a solution method based on Markov chain Monte Carlo (MCMC). In the framework of MCMC, the Gibbs sampling algorithm is introduced by deriving the fully conditional distribution of all parameters and hyperparameters.

B. Gibbs Sampling

The Gibbs sampling algorithm is used to iterate the above method. The idea of Gibbs sampling is to update the variables sequentially in each iteration. For each variable, all other variables are fixed at the current value, and samples are taken from the cloth. Sampling the factor matrix is to obtain the dependence between the observed value $d_{i,j,t}$ and the hyperparameters. After the Gibbs sampling algorithm reaches a stationary state, all missing observations can be approximated from MCMC.

III. Method Developed in This Study

This section mainly introduces our proposed HTR-BTTF method. We define tensor data $D \in \mathbb{R}^{t\times m\times n}$ again, where m and n represent the spatial side length of the study area, respectively, and t represents the time-series length. The HTR-BTTF method includes the following five steps.

A. Hilbert Curve to Dispersion Missing Areas

There is an obvious aggregation phenomenon of cloud cover in remote sensing data. The traditional method is prone to distortion after interpolation. Using the Hilbert curve to reduce dimensionality can improve the spatial heterogeneity of the cloud cover area. The Hilbert curve is a kind of spatial filling curve, which is mainly applied to spatial geographic data encoding index. It can linearly penetrate every discrete element in two or three dimensions. Also, through only once, Hilbert curves can map data in a higher dimensional space that is not well ordered into a 1-D space, where adjacent objects are stored in proximity. To describe a Hilbert curve in two dimensions, the order is the only parameter. Set the length of the side as d, and then, $n = \log_2 d$.

We use the Hilbert filling curve to fill the study area, and the missing observations can be effectively dispersed. The Hilbert curve is only suitable for the study of side length of 2^{ord} (ord is the order of the Hilbert curve) region, but the actual sample does not meet this condition; therefore, the Hilbert filling curve with the minimum order of $\lceil \log_2 \max(m, n)\rceil$ is required. We can place the study area in the center of the filling curve, which can fill an area of any size. As shown in Fig. 1, the white squares represent missing observations. It can be seen that the missing observations in the tensor are effectively dispersed, and the tensor data after the rearrangement become $D_h \in \mathbb{R}^{t\times S}$, where $S = m \times n$.

B. Rearrangement Temporal Dimension

LST data need to meet the observation number of integer years, i.e., $t = Y \times T$, where T is the period and Y is the number of observation years. As shown in Fig. 1, the new tensor data are rewritten as $D_{\text{HTR}} \in \mathbb{R}^{T\times Y\times S}$. After the rearrangement of temporal dimension, the periodic properties of LST data in time series are extracted.

C. Restack Tensor

As we all know, the BTF method is relatively time-consuming when processing large-size data to calculate

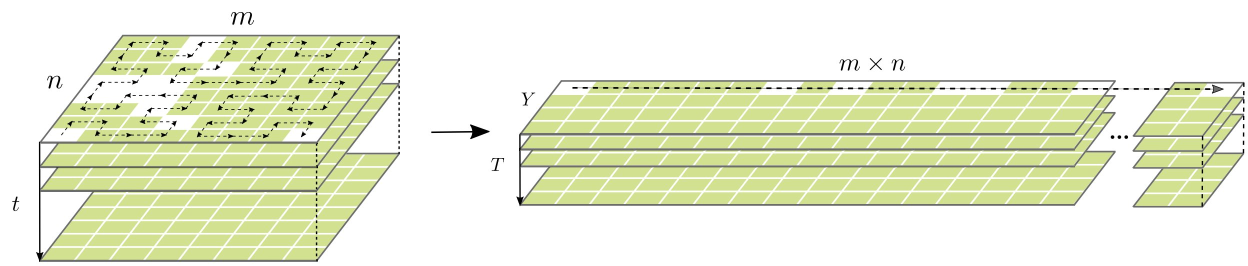


Fig. 1. Schematic of tensor rearrangement, which is divided into two steps. Step 1: Spatial rearrangement: rearrange the spatial data using Hilbert curve and m and n are the side lengths of the data. Step 2: Temporal rearrangement: Y is the year, T is the number of observations in a year, and $t = Y \times T$.

the posterior distribution and Gibbs sampling, but its data recovery ability often conceals this shortcoming. In processing time-series remote sensing image data, the time-series length and the side length of the image are usually relatively large. If the spatial segmentation method is used for regional interpolation, it will produce a very obvious edge boundary phenomenon. Thus, in order to solve this problem, we further improve the method obtained in the previous step, that is, set an adjustable parameter *block* (*block* is equal to one by default). We propose to restack the rearrangement tensor, as shown in Fig. 2(a).

When *block* is equal to one, adjacent pixels on the filled curve are absolutely adjacent in space, which is very helpful for spatial interpolation. When the *block* is equal to two, the original filling curve branches out n pieces to fill one-half of the space, as shown in Fig. 2(b), and so on. Setting the *block* in a certain range can ensure that the adjacent pixels of the filling curve of each branch bar have a certain spatial correlation. The length of the filled curve is shortened and satisfies the proximity relationship. Experiments have shown that this method can shorten the running time and does not produce higher errors. This will be described in detail in Section III-D.

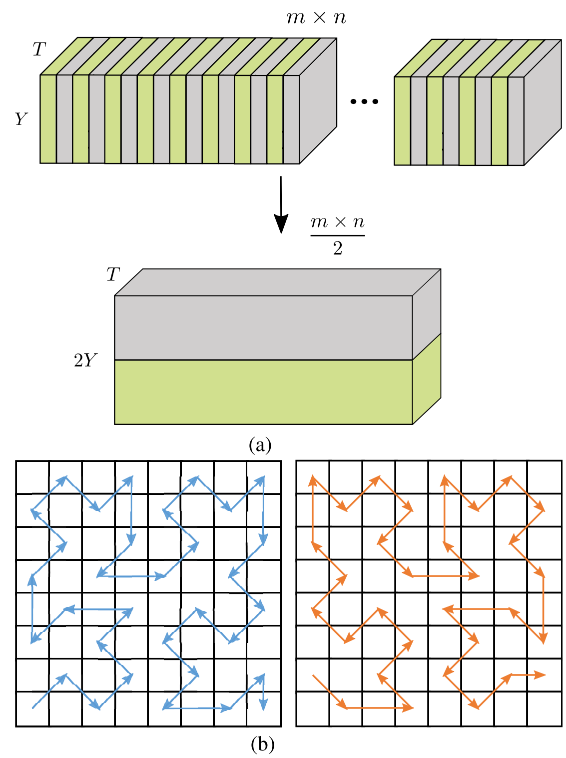


Fig. 2. Schematic of tensor restack, and parameter *block* in both figures is two. (a) Rearrangement tensor, which is sampled every *block* in the spatial dimension direction and restacked into a new tensor. The tensor dimension changes from $T \times Y \times mn$ to $T \times 2Y \times mn/2$. (b) Filling curve of the restacked tensor changes. (a) Schematic of tensor restack. (b) Two branches of Hilbert curve after tensor restack.

D. BTTF Method Interpolation

Input the rearrangement tensor into the BTTF method for interpolation. The difference from it is that the Y and T dimensions can be initialized as two temporal factor matrices.

E. Restore Tensor

After the interpolation, we need to perform the inverse process of Sections III-A–III-C to restore the tensor data.

The early stopping mechanism is used in the training stage of the method. We define that when the root-mean-square error (RMSE) decrease of nearly 100 iterations is less than a specified threshold, the burn-in period stops, and then, the average value of g times Gibbs sampling results is taken as the final experimental result.

From the perspective of Bayesian theory, the BTF method does not regard the parameter as a fixed value, but as a random variable subject to a certain distribution, and instead estimates the parameters of the distribution. This method can reflect the process of interaction within the data and reduce the sensitivity of the method to abnormal data. HTR-BTTF uses the VAR to process the time factor matrix to improve the method's interpolation accuracy for the time series. Moreover, the HTR-BTTF interpolation method is not pixel-by-pixel interpolation. It considers the global value distribution in each iteration, and it is based on the MCMC estimation, which has nothing to do with the distribution of the initial parameters and also means that we do not need the initial state of the relationship parameters before use.

IV. Experiments

A. Data and Preprocessing

1) Experimental Data: MOD11A2 (V006) data from the Terra satellite were selected in this research. These daily MODIS LST data at a 1-km spatial resolution have data quality information for each pixel and time resolution is eight

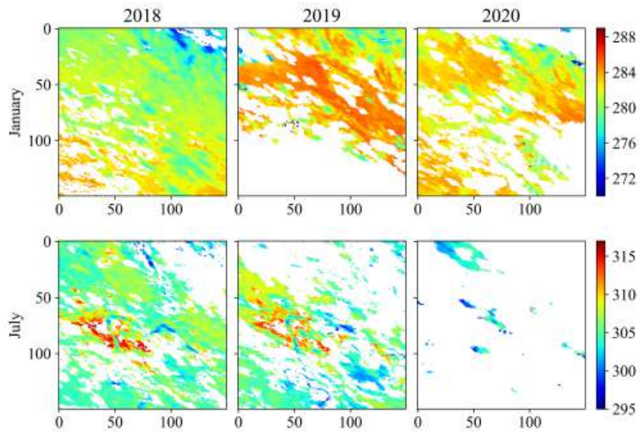


Fig. 3. Wuhan LST images after data preprocessing. The top three images are the first images of each year, and the bottom three images are the 25th images of each year.

TABLE I

MRR OF ARTIFICIAL CLOUD COVER AND THE CORRESPONDING CLASSIFICATION USED IN THE LST SIMULATION LEVEL OF THE STUDY AREA SURFACE (%)

Size	A	B	C	D	E
MRR	5∼15	20∼35	45∼60	60∼80	80∼90

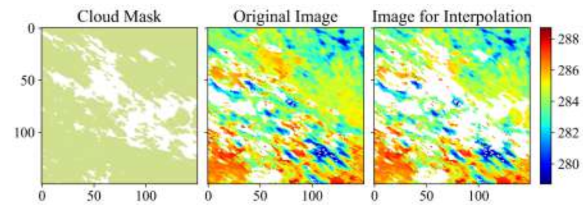


Fig. 4. Place size B cloud cover on the test image of Wuhan. The white pixels in the cloud mask image are missing observation data.

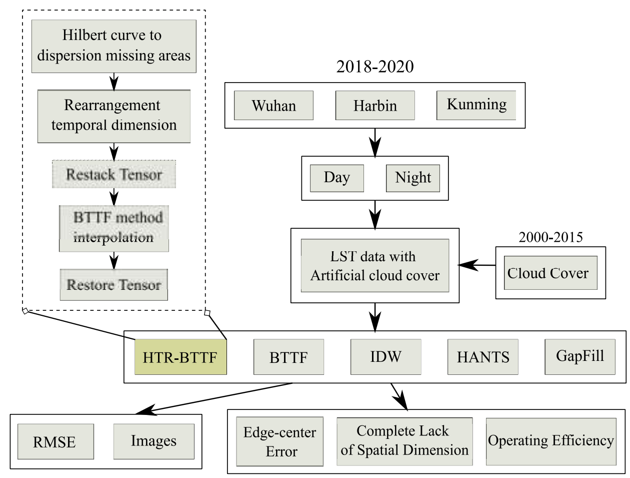


Fig. 5. Flowchart of the simulation study and the process of HTR-BTTF method.

days, so 46 images are available every year. According to theory, eight-day LST synthesis is available [35], which is obtained from all clear sky observations for a given pixel in an eight-day period that is averaging. The proportion of missing pixels in everyday LST images is high, which often makes it impractical. Therefore, eight-day images are more convenient to use in many cases [36], [37], and the synthesis of eight-day images reduces storage space requirements and calculation time.

The impact of surface temperature on the atmospheric circulation leads to different climates. We choose four different climate regions as the result of comparison, Wuhan City with subtropical monsoon climate, Harbin City with temperate monsoon climate, and Kunming City with subtropical plateau monsoon. From 2018 to 2020, we collected a total of 138 images. These areas consist of a 150 × 150 rectangular array.

2) Preprocessing: LST data and its pixel quality and other information are stored in Hierarchical Data Format (HDF) files with local attributes of scientific datasets (SDSs), where we can find the daytime LST data and the corresponding quality assurance (QA) layer. According to the data quality tags contained in the MODIS LST data, the pixels with clouds, an average emissivity error > 0.04, and an average LST error > 2K were marked as null pixels, the pixels with invalid LSTs. LSTs are given in kelvin degrees [38]–[40]. Fig. 3 shows the January and July images of Wuhan from 2018 to 2020. Obviously, a large number of pixels have lost LST data, resulting in data unavailability at all.

B. Experiment Design

1) Data Preparation: The experiment used day and night LST image data of three study areas and added cloud covers with different missing rates. It is difficult to find high-quality images with no data loss in the study area, so we chose the images with the lowest missing rate in the cold season (January and February) and warm season (July and August) each year. A total of six images were added with cloud cover in each study area.

In order to make the artificial cloud cover closer to real cloud cover, we selected the mask data of the real cloud cover at different missing rates in each study area during the 15 years from 2000 to 2015 and classify the missing rate range (MRR), as shown in Table I. A and B were defined as small missing rates, C and D as medium missing rates, and E as large missing rates. Fig. 4 shows our actual process.

2) Model Comparison: For the HTR-BTTF method, we set the same experimental parameters for the three research areas, the rank is set to 40, the threshold for early stopping is set to 0.05, the number of Gibbs sampling g is set to 200, and *block* is set to one. It is worth noting that the HTR-BTTF method only needs to know the period of the data in advance and does not require other *a priori* parameters.

In addition, we compare four interpolation methods.

1) The original BTTF method does not add tensor rearrangement, and we set it with the same parameters as the method we proposed.
2) The spatio-based interpolation method IDW uses the distance between the interpolation point and the sample point as the weight to perform a weighted average.

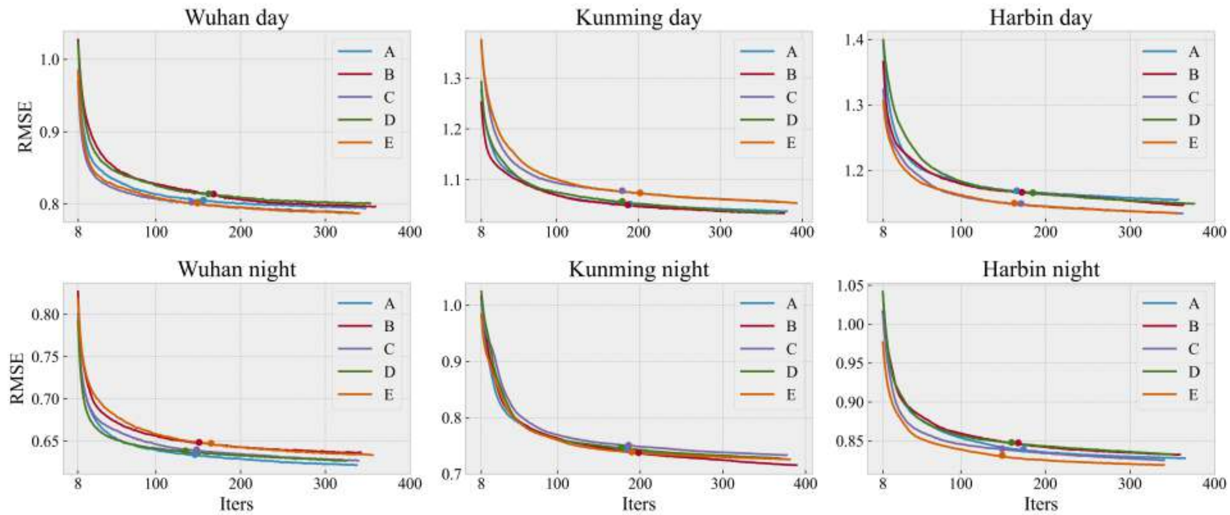


Fig. 6. RMSE change trend of the day and night interpolation process after eight iterations in Wuhan, Kunming, and Harbin. The curves with different colors represent the five types of missing rates, and the color points represent the starting point of Gibbs iterations.

The closer the sample point is to the interpolation point, the greater is the given weight. IDW obtains the interpolation unit by averaging the value of each sampling point in the neighboring area [41].

3) The temporal-based interpolation method HANTS [15] was originally developed for processing time series of noise remote sensing data, but it has been used to interpolate invalid data in eight-day MODIS LST images [2]. HANTS is an improved method based on the Fourier transform method. Its basic principle is to use the average value of the time series and several important harmonics to perform the iterative solution process of the least square fitting of the time-series signal. In this article, HANTS is used to interpolate pixel by pixel. Suitable parameters are estimated in advance. The parameters of the experimental results are the best results through continuous attempts.
4) The temporal–spatial-based interpolation method Gap-Fill [18] selects the image to be interpolated to become a prediction set and sorts it based on the numerical comparison between all the images in the prediction set. For all the observation values in the spatial position of the target value in the prediction set image, it determines which empirical quantile the value corresponds with respect to all values of the image. The average of the empirical quantile levels obtained from all images is taken and used as the target quantile level, and then, quantile regression is used to fill the gap [19]. The R language package is also provided [18], which makes the method easier to implement.

3) Evaluating Indicator: At present, the RMSE is generally used to reflect the estimation sensitivity and extreme effect of sample points. Moreover, if the RMSE value is larger, the accuracy of the interpolation method will be worse, but if on the contrary, it will be better. Based on the experiment in this article, the following evaluation indexes are given. Also, we calculate the RMSE for each image

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{11}$$

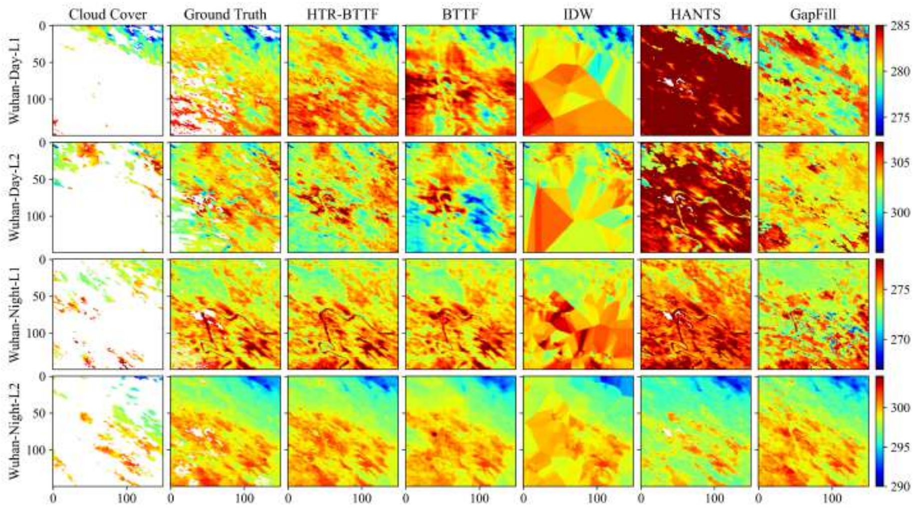
where y_i is the true value, $\hat{y}_i$ is the predicted value, and N is the number of missing pixels of each image.

In addition, we also compared the distortion of the image after interpolation and the operating efficiency. Fig. 5 shows the flowchart of the simulation study and the process of HTR-BTTF method.

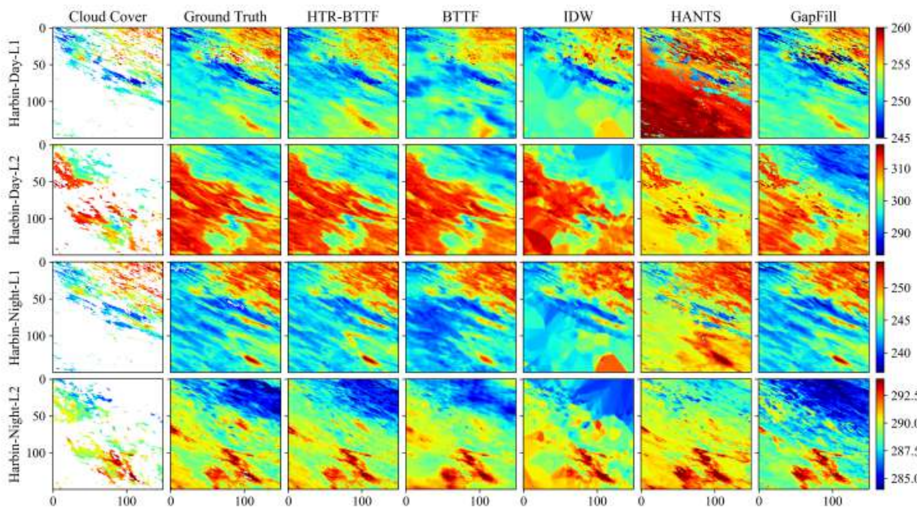
C. Result

The traditional iterative process of Bayesian factorization needs to specify burn iterations, but the early stopping method is more effective. Fig. 6 shows the fitting process of the three regions when the block is one. It can be seen that the method only needs to burn 100–200 iterations, and more iterations will not significantly help the improvement of method accuracy.

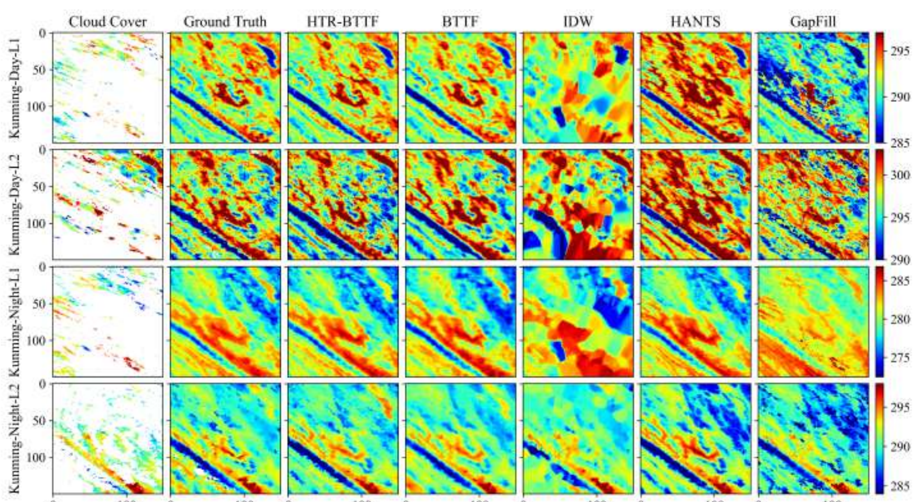
Table II shows the RMSE calculated under different sizes of cloud cover in the three study areas, where a total of six images (three images for cold seasons and three images for warm seasons) and two scenes of day and night are set for each cloud cover size. (Some words and methods in the table use abbreviations, HTR stands for HTR-BTTF method, BF stands for BTTF method, HS stands for HANTS method, GF stands for GapFill method, S stands for Size, and L stands for the images number to be interpolated). It can be seen that the HTR-BTTF method contributes the best RMSE in most of the experiments, especially when the cloud cover has the largest size; on average, the RMSE of the HTR-BTTF method was 17.2% lower than that of BTTF, 41.9% lower than that of IDW, 56.3% lower than that of IDW, and 54.9% lower than that of Gapfill. This proves that the HTR-BTTF method can complete the LST image interpolation of different cloud cover sizes, different climates, and different seasons without many prior parameters.



(a)



(b)



(c)

Fig. 7. Interpolation images of different methods in the cold and warm season layers in the first year. (a) Wuhan. (b) Harbin. (c) Kunming. Cloud cover size: E.

TABLE II

RMSE VALUES OBTAINED AFTER INTERPOLATION OF CLOUD COVER DATA OF DIFFERENT SIZES USING HTR-BTTF, BTTF, IDW, HANTS, AND GAPFILL METHODS

method		HTR	BF	IDW	HS	GF	HTR	BF	IDW	HS	GF	HTR	BF	IDW	HS	GF	HTR	BF	IDW	HS	GF	HTR	BF	IDW	HS	GF	HTR	BF	IDW	HS	GF
S & L		Wuhan Day					Wuhan Night					Harbin Day					Harbin Night					Kunming Day					Kunming Night				
A	L1	0.73	0.78	**0.70**	5.17	1.11	0.61	**0.58**	1.00	2.68	1.79	0.83	0.94	**0.69**	7.03	3.23	**0.77**	0.93	1.09	4.57	2.52	**0.74**	0.81	1.24	2.13	3.85	**0.34**	0.43	0.53	1.38	1.73
	L2	1.07	0.94	**0.91**	4.54	1.98	0.50	0.56	**0.49**	1.10	2.04	**1.31**	**1.31**	1.53	6.38	5.07	0.61	0.71	**0.52**	1.08	1.88	1.81	1.74	**1.60**	2.55	3.41	**0.64**	0.70	0.75	0.89	1.67
	L3	**0.77**	0.82	0.96	1.64	1.90	**0.42**	0.52	0.83	1.43	2.09	1.23	0.95	**0.87**	4.19	4.27	**0.85**	0.89	1.20	4.52	2.63	**0.53**	0.66	1.51	2.13	1.23	**0.51**	0.55	0.90	1.66	0.75
	L4	**0.70**	0.83	1.02	1.04	1.57	0.67	**0.55**	0.63	1.77	0.86	0.86	0.82	0.71	4.68	**0.71**	**0.51**	0.60	0.61	1.14	2.32	**1.53**	1.53	2.21	1.64	3.69	1.01	0.92	**0.90**	1.94	3.44
	L5	**0.51**	0.57	0.75	3.82	1.04	0.73	**0.70**	0.83	2.29	3.00	**1.31**	1.35	1.49	8.60	4.02	**0.80**	0.87	1.28	7.38	1.39	**0.71**	0.97	1.91	1.30	1.08	**0.43**	0.60	1.03	0.79	0.73
	L6	**0.60**	0.73	0.89	1.26	1.43	**0.35**	0.62	0.71	2.25	0.86	**0.75**	1.03	0.87	3.28	1.43	**0.95**	2.25	1.34	1.74	1.64	**0.84**	0.93	1.19	2.20	3.31	0.47	**0.44**	0.53	1.25	2.00
B	L1	**0.74**	0.76	0.97	5.57	1.39	**0.64**	0.68	0.97	2.66	2.07	**0.85**	0.99	0.94	5.89	3.31	**0.80**	1.06	1.13	3.48	3.17	**0.63**	0.74	1.16	1.70	1.70	**0.34**	0.49	0.65	1.06	1.40
	L2	1.03	**1.02**	1.39	3.71	1.20	**0.54**	0.60	0.64	1.55	2.35	**1.48**	1.67	2.20	4.73	4.61	0.83	0.80	**0.76**	1.34	1.62	1.87	**1.81**	1.85	2.58	2.58	0.91	**0.86**	1.12	1.24	1.88
	L3	**0.73**	0.80	0.97	1.98	1.58	**0.48**	0.50	0.92	1.58	1.74	1.02	**0.94**	1.06	2.31	2.21	**0.65**	0.67	0.97	4.91	1.18	**0.56**	0.73	1.40	2.08	2.08	**0.46**	0.51	0.87	1.51	0.81
	L4	**0.68**	0.80	1.43	0.97	1.68	**0.55**	0.55	0.69	1.76	0.84	**0.71**	0.94	0.77	4.43	0.80	0.66	0.63	**0.61**	1.49	2.76	1.33	**1.32**	1.42	1.76	1.76	1.05	0.92	**0.83**	1.67	2.80
	L5	**0.71**	0.78	0.90	3.55	1.59	**0.81**	0.93	1.23	2.13	2.93	1.08	1.16	**1.07**	9.30	3.70	**0.70**	0.82	1.08	8.47	1.35	**0.73**	0.88	1.17	1.83	1.83	**0.47**	0.51	0.59	1.20	1.07
	L6	0.92	**0.89**	1.08	1.42	1.65	**0.48**	0.49	0.49	2.34	1.07	0.96	1.03	0.89	3.52	**0.86**	**1.02**	1.26	1.08	2.43	2.55	**0.82**	0.91	1.69	1.95	1.95	**0.46**	**0.46**	0.95	1.43	2.74
C	L1	**0.78**	0.86	1.11	5.28	1.16	**0.70**	0.72	1.35	2.77	2.20	**1.14**	1.22	2.95	2.93	3.78	**1.14**	1.22	2.95	2.93	3.78	**0.95**	0.96	2.48	1.41	3.27	**0.45**	0.59	1.49	0.99	1.27
	L2	1.12	**0.97**	1.14	4.10	1.48	0.67	**0.53**	0.67	0.81	1.72	0.93	**0.85**	0.78	1.59	1.72	0.83	0.85	**0.78**	1.59	1.72	1.87	**1.73**	2.07	2.85	3.50	0.90	**0.84**	1.13	1.08	1.65
	L3	**0.64**	0.77	0.96	1.62	1.88	**0.44**	0.52	0.86	1.54	1.48	**0.86**	0.92	1.60	5.21	4.90	**0.86**	0.92	1.60	5.21	2.90	**0.61**	0.74	1.29	2.09	1.77	**0.44**	0.58	0.84	1.50	1.04
	L4	**0.77**	0.89	1.70	0.91	1.67	**0.59**	0.59	0.90	1.63	0.82	**0.62**	0.65	0.79	1.35	1.57	**0.62**	0.65	0.79	1.35	2.57	**1.44**	1.45	2.47	2.05	3.75	1.08	**0.91**	1.16	1.58	2.69
	L5	**0.75**	0.85	1.15	3.75	1.79	**0.82**	1.08	1.60	2.25	2.90	**0.82**	0.89	1.42	8.29	5.30	**0.82**	0.89	1.42	8.29	1.30	**0.72**	0.88	1.39	1.91	1.18	**0.54**	0.57	0.65	1.24	1.28
	L6	**0.85**	0.88	1.46	1.26	1.34	**0.39**	0.53	0.78	2.35	0.97	**1.23**	1.30	1.28	2.48	4.30	**1.23**	1.30	1.28	2.48	2.30	**0.84**	0.84	1.85	2.15	3.61	**0.47**	0.51	0.99	1.24	1.85
D	L1	0.84	**0.79**	1.20	5.31	1.08	**0.64**	0.68	1.22	2.78	2.17	**1.57**	1.68	2.23	6.39	3.32	**0.97**	1.42	2.82	4.40	2.75	**0.74**	0.78	2.05	1.98	3.63	**0.37**	0.50	1.15	1.20	1.63
	L2	1.16	**1.02**	1.48	3.69	1.53	0.57	**0.53**	0.67	1.14	2.17	**1.51**	1.79	2.71	4.05	5.95	**0.96**	0.97	1.13	1.80	1.95	1.90	**1.81**	2.80	2.83	3.51	**0.78**	0.86	1.34	1.08	1.99
	L3	**0.82**	0.92	1.75	2.02	1.55	0.59	**0.58**	1.38	1.66	1.71	**1.20**	**1.20**	1.45	2.69	2.74	**0.81**	0.96	1.39	4.73	1.68	**0.67**	0.77	2.58	2.14	1.76	**0.49**	0.68	1.64	1.59	1.06
	L4	**0.88**	1.18	1.66	0.98	1.88	**0.59**	0.61	0.99	1.63	0.80	**1.01**	1.10	1.41	4.57	1.55	**0.73**	**0.73**	1.21	1.52	2.71	1.51	**1.48**	2.83	2.32	3.35	1.49	1.17	2.30	1.49	2.58
	L5	**0.98**	1.68	1.40	3.57	1.78	**1.21**	1.74	1.75	2.14	2.89	1.23	**1.21**	1.31	9.00	3.12	**0.79**	0.87	1.51	8.77	1.69	**0.77**	0.94	1.55	1.83	1.13	**0.54**	0.61	0.76	1.17	1.17
	L6	**1.04**	1.46	1.69	1.35	1.52	**0.47**	0.75	1.71	2.19	1.16	**1.04**	1.12	1.30	3.42	1.87	**1.23**	1.40	1.73	2.34	2.36	**0.92**	0.92	2.60	2.03	3.70	0.70	**0.53**	1.73	1.37	2.20
E	L1	**0.92**	1.38	1.42	5.29	2.16	**1.07**	1.33	1.52	2.74	2.21	**1.41**	1.48	1.41	6.76	3.19	**0.78**	1.35	1.87	4.40	2.86	**0.83**	0.94	2.56	2.05	3.79	**0.42**	0.54	1.51	1.31	1.81
	L2	**1.27**	1.56	1.66	3.76	1.61	**0.99**	1.00	1.22	1.06	1.88	**1.67**	1.68	2.97	4.44	5.20	**1.06**	1.09	1.12	1.72	1.79	**1.33**	1.93	3.77	2.94	3.49	**0.89**	0.91	2.95	1.08	1.82
	L3	**0.82**	1.09	1.62	1.85	1.77	**0.53**	0.58	1.27	1.57	1.87	**1.02**	1.06	1.24	3.57	3.27	**0.86**	0.90	1.53	4.85	2.09	**0.68**	0.78	2.27	2.13	1.81	**0.51**	0.55	1.41	1.54	1.30
	L4	**0.78**	0.94	1.70	0.93	2.69	**0.54**	0.71	0.88	1.53	0.75	**1.05**	2.00	1.65	4.56	1.60	**1.25**	1.41	1.53	1.48	2.67	**1.50**	1.56	2.63	2.14	3.74	**1.08**	1.20	1.69	1.63	2.77
	L5	**1.11**	1.32	1.60	3.65	2.28	**1.09**	1.83	1.41	2.16	2.90	**1.13**	1.52	1.29	9.08	3.02	**1.01**	1.03	1.64	8.86	1.77	**0.95**	1.17	2.91	2.11	1.30	**0.64**	1.03	1.84	1.41	1.40
	L6	**0.96**	1.67	2.07	1.23	1.73	**0.52**	0.87	1.14	2.23	1.09	**1.12**	1.14	1.29	3.54	2.12	**1.39**	1.54	2.02	2.43	2.23	**1.12**	1.34	2.43	2.09	3.55	**0.80**	0.85	1.46	1.31	2.09

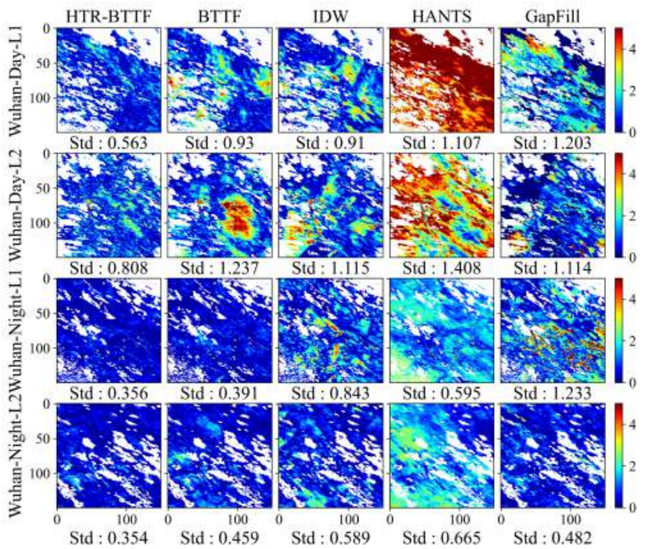


Fig. 8. Interpolation error calculated pixel by pixel and the Std for HTR-BTTF, BTTF, IDW, HANTS, and GapFill.

The interpolation effect of the HTR-BTTF method and the BTTF method is basically the same for small and medium cloud cover sizes. However, when faced with a large-scale cloud cover, the BTTF method lacks too many observations on a single layer, resulting in poor interpolation effect, but the HTR-BTTF method can solve this problem well. The IDW method relies on the spatial similarity of surrounding pixel values. In the case of a small cloud cover size, it shows a lower RMSE, but when faced with a large cloud cover, IDW cannot obtain observations from the adjacent areas, resulting in ineffective work. The HANTS method has problems when faced with very extreme data. In the experimental data, a small part of the pixels will be completely lost in the time sequence, resulting in the inability to perform the effective interpolation. The HANTS method has many adjustable parameters, and a fixed set of parameters cannot satisfy all pixels at the same time, which is also the reason for the poor effect of HANTS interpolation. The GapFill method relies on the adjacent relationship between spatial and temporal. However, the interpolation result is unstable. When the adjacent effective information is insufficient, a larger error will be caused. The GapFill interpolation process is very time-consuming, and the running time of GapFill interpolation will continue to increase as the missing rate increases.

Overall, HTR-BTTF outperforms the BTTF, IDW, Hants, and GapFill regardless of the cloud size for all two variables, LST Day and LST Night. Although the difference in the values of the RMSE calculated under the E-size cloud cover of the five methods is not obvious, the RMSE index does not fully reflect the interpolation effect of the method. Fig. 7 shows the interpolation results of the two images in the first year of the three study areas of the E size cloud cover. The interpolation results using the HTR-BTTF method are more natural in texture and visually very similar to the real temperature distribution. The image after interpolation by the BTTF method will show obvious blurring (Wuhan-Day-L1 and Wuhan-Day-L2) and sometimes overestimate or underestimate the temperature value in some areas (Harbin-night-L1 and Harbin-night-L1 and Harbin-Day-L2), lack of spatial heterogeneity. IDW, Hants, and GapFill methods are available when dealing with small-scale cloud cover, but they are not suitable for interpolating more than 80% of the missing in a single image.

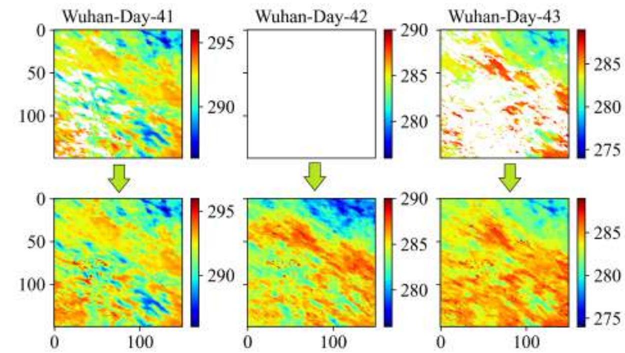


Fig. 9. Interpolation results of the 41st, 42nd, and 43th images of the HTR-BTTF method in Wuhan during the daytime, the 42nd image is completely missing in space.

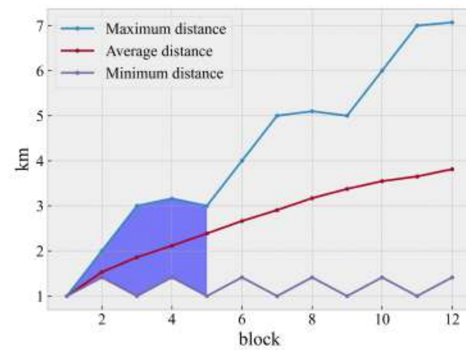


Fig. 10. Maximum, minimum, and mean value of the distance between the adjacent pixels on the Hilbert branch curve in different *block*.

TABLE III

COMPARISON OF FIVE METHODS TO INTERPOLATE WUHAN CITY'S DAY TIME (MIN)

Size	HTR-BTTF	BTTF	IDW	HANTS	GapFill
A	48.63	7.39	0.25	0.87	1501.92
B	47.21	7.15	0.24	0.87	1520.06
C	46.45	7.61	0.24	0.87	1548.75
D	49.69	7.94	0.24	0.87	1561.80
E	47.54	7.27	0.24	0.84	1586.90

V. DISCUSSION

A. Edge-Center Error

As is well known, current interpolation methods are mostly based on the neighboring similarity principle, so when there is a lack of large area, the RMSE value is greater in the border pixels than in pixels located in the center of the cloud cover [19]. If a missing area would lead to an error stack, a follow-up study could easily lead to wrong conclusions.

TABLE IV

RMSE Value Obtained After the HTR-BTTF Method With Different *Block* Interpolates the Cloud Cover Data of Different Sizes and the Comparison of the Running Time

	block		1	2	3	4	5	1	2	3	4	5
	S & L		Day					Night				
Wuhan	A	L3	0.77	0.76	0.82	0.87	0.79	0.42	0.42	0.43	0.45	0.46
		L4	0.70	0.67	0.71	0.71	0.72	0.67	0.66	0.64	0.69	0.64
	B	L3	0.73	0.74	0.75	0.75	0.79	0.48	0.46	0.47	0.48	0.49
		L4	0.68	0.72	0.71	0.72	0.71	0.55	0.55	0.55	0.56	0.56
	C	L3	0.64	0.65	0.67	0.68	0.66	0.44	0.46	0.47	0.48	0.51
		L4	0.77	0.80	0.82	0.83	0.79	0.59	0.60	0.61	0.64	0.64
	D	L3	0.82	0.85	0.85	0.87	0.84	0.59	0.55	0.64	0.63	0.59
		L4	0.88	0.85	0.81	0.96	0.86	0.59	0.59	0.59	0.60	0.67
	E	L3	0.82	0.81	0.84	0.94	0.83	0.53	0.54	0.55	0.59	0.61
		L4	0.78	0.81	0.86	0.87	0.84	0.54	0.56	0.60	0.58	0.58
	block		1	2	3	4	5	1	2	3	4	5
	S & L		Day					Night				
Harbin	A	L3	1.23	1.25	1.25	1.26	1.24	0.85	0.90	0.90	0.94	0.90
		L4	0.86	1.02	1.09	1.02	0.97	0.51	0.55	0.57	0.58	0.64
	B	L3	1.02	1.10	1.10	1.15	1.06	0.65	0.68	0.85	0.71	0.79
		L4	0.71	0.75	0.84	0.83	0.86	0.66	0.67	0.64	0.68	0.65
	C	L3	0.86	0.88	0.88	0.93	0.95	0.86	0.87	1.01	1.02	1.00
		L4	0.62	0.62	0.66	0.71	0.76	0.62	0.65	0.64	0.68	0.72
	D	L3	1.20	1.24	1.32	1.36	1.37	0.81	0.93	0.99	0.90	0.93
		L4	1.01	1.13	1.06	1.15	1.11	0.73	0.71	0.80	0.82	0.76
	E	L3	1.02	1.14	1.20	1.29	1.33	0.86	0.93	0.84	1.02	1.08
		L4	1.05	1.15	1.32	1.27	1.29	1.25	1.13	0.94	1.25	1.30
	block		1	2	3	4	5	1	2	3	4	5
	S & L		Day					Night				
Kunming	A	L3	0.53	0.51	0.54	0.51	0.55	0.51	0.50	0.50	0.52	0.56
		L4	1.53	1.53	1.54	1.57	1.58	1.01	1.08	1.02	1.02	1.11
	B	L3	0.56	0.56	0.57	0.59	0.58	0.46	0.45	0.48	0.50	0.54
		L4	1.33	1.36	1.35	1.34	1.35	1.05	1.04	1.04	1.08	1.09
	C	L3	0.61	0.65	0.65	0.64	0.63	0.44	0.46	0.49	0.50	0.51
		L4	1.44	1.56	1.60	1.52	1.47	1.08	1.13	1.12	1.10	1.13
	D	L3	0.67	0.63	0.55	0.65	0.60	0.49	0.50	0.53	0.51	0.55
		L4	1.51	1.58	1.62	1.64	1.67	1.49	1.50	1.41	1.51	1.51
	E	L3	0.68	0.69	0.65	0.70	0.71	0.51	0.52	0.55	0.54	0.52
		L4	1.50	1.53	1.66	1.70	1.72	1.08	1.11	1.16	1.18	1.18
Time/(min)			47.90	26.31	17.13	14.03	12.17					

However, the method we proposed in this article can solve this problem well. Taking Wuhan as an example, Fig. 8 shows the pixel error of the five methods after interpolation and the standard deviation (Std). As can be seen in the figure, the error distribution in the HTR-BTTF method is relatively uniform, which will not cause the interpolated data to be unavailable. The Std of the error using HTR-BTTF interpolation is the smallest. The effect of interpolation at night is better than that in daytime. The BTTF method and HTR-BTTF have basically the same interpolation effect for night data, but the error of the day data at the center of the cloud cover is very obvious. This can prove that it is effective to use the Hilbert space-filling curve to disperse the large cloud cover.

B. Complete Lack of Spatial Dimension

In this experiment, the 42nd image data of Wuhan are completely missing spatial day data. We cannot judge whether HTR-BTTF can effectively deal with completely missing images in space. Thus, we indirectly judge the reliability of its interpolation, as shown in Fig. 9 with the 41st and 43rd images, the images before and after the time sequence of the 42nd, respectively. It can be seen that the maximum data range of the three images is gradually reduced. This is in line with the actual situation. From November to December, Wuhan enters winter, the climate becomes colder, and the surface temperature will drop. Second, by observing the image, it can be seen that there is no blurring. Therefore, we can conclude that without the support of other data sources, HTR-BTTF can well complete the complete lack of space.

C. Weigh the Error and Operating Efficiency

As for computing time, Table III shows the running times required for processing 138 images in Wuhan and different sizes of artificial clouds with all methods on a Windows PC with an Intel Core i7-10700 and 32 GB of RAM. The running time of HTR-BTTF method in the table includes the total time of tensor rearrangement, interpolation, and tensor restoration. In addition to the interpolation process, other operations are not very time-consuming. The interpolation time of the other four methods includes the data preprocessing process, and the

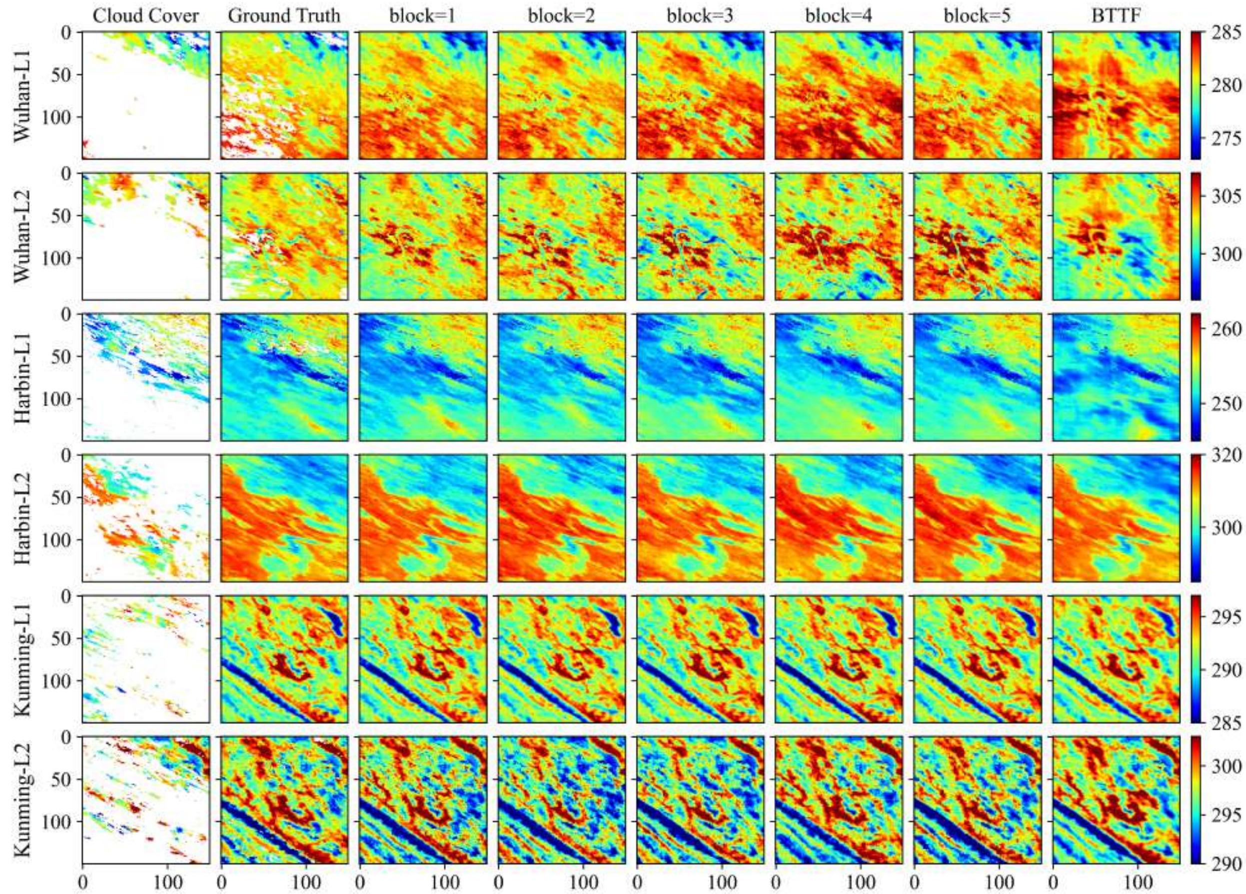


Fig. 11. Comparison of interpolation results between HTR-BTTF with different *block* and BTTF methods. Cloud cover size: E.

time to export to tag image file format (TIFF) is not included. It is acceptable for the HTR-BTTF method to process an image in an average of 20 s, but more efficient improved methods are worth exploring.

Different *block*s have different effects on the Hilbert fill curve. The maximum, minimum, and average values of the distance between the adjacent pixels on the Hilbert branch curve in terms of spatial geographic location information vary with the *block*, as shown in Fig. 10. When the *block* is five, the longest distance in space between the adjacent points on the branch Hilbert curve is three. Table IV shows the RMSE results calculated for three study areas and different cloud covers with different *block*. Only the third and fourth imputation layers are shown in the table (i.e., the cold and warm seasons in 2019). It can be seen that in the three study areas, as the *block* increases, the RMSE of most experimental layers fluctuates within an acceptable range, and it is still better than other comparative methods under the cloud cover with a large missing rate. From the last line in the table, we can see that the operating efficiency is significantly improved. When the *block* is equal to five, the running time is reduced to one-quarter of the original, and one layer can be interpolated in about 5 s on average.

Taking daytime as an example, Fig. 11 shows the interpolation images of different *block* and BTTF method in three study areas with the E-size cloud cover. It can be seen that setting different *block* has little effect on the interpolation results, and there is no interpolated image distortion such as the BTTF method, but it is undeniable that the effect of *block* equal to temporary is better. Therefore, in terms of the balance of accuracy and efficiency, different tasks should have different choices. The results of our proposed method in terms of usability, robustness, and interpolation efficiency are worthy of recognition.

In order to test the interpolation efficiency and accuracy of the HTR-BTTF method in different data scales, we used bilinear interpolation to spatially resampling the daytime surface temperature data in Wuhan, expanded the data size to 200×200, 300×300, 400×400, and 500×500, and increased the time span to ten years. The experiment does not add artificial cloud cover, and the final indicator is determined by the iterative stable RMSE. The experimental setting block is equal to five, and early stopping is also used in this experiment. The results are shown in Table V. It can be seen that the stable RMSE results of different image sizes can be maintained in a very small range, which can prove the ability of our proposed HTR-BTTF method to learn the data distribution and the effectiveness of tensor overlap. The larger the spatial size has a greater impact on the interpolation time than the larger the time span, so we believe that HTR-BTTF is suitable for the

TABLE V
HTR-BTTF METHOD IS APPLIED TO THE TIME AND RMSE OF DIFFERENT SPACE SIZES AND DIFFERENT TIME-SERIES LENGTHS (MIN)

Space size		200×200		300×300		400×400		500×500	
Time Span	Time series length	RMSE	Time	RMSE	Time	RMSE	Time	RMSE	Time
3 years	138	0.85	20.41	0.87	29.73	0.95	79.43	0.87	127.78
5 years	230	0.91	25.13	0.92	37.13	0.89	101.93	0.96	167.83
8 years	368	0.87	31.38	0.95	46.19	0.86	132.11	1.06	225.10
10 years	460	1.01	38.25	0.96	58.92	0.98	162.20	0.99	316.22

interpolation of remote sensing data in a small spatial range and long time series.

VI. CONCLUSION

In this study, a method called the HTR-BTTF method that integrates spatiotemporal information. The accuracy of the HTR-BTTF method was compared with the accuracies of four Interpolation methods. Experiments prove that the performance of our proposed method is better than the BTTF, IDW, HANTS, and GapFill methods. HTR-BTTF method uses the Hilbert space-filling curve to change the distribution of missing values from concentrated to scattered, which improves spatial heterogeneity. The reconstruction in the time dimension ensures the relevance and periodicity of the time series, and the Bayesian factorization method is used to fit the distribution of the data for interpolation of the data.

In general, the HTR-BTTF method is competitive for several reasons: 1) it performs interpolation based on the overall data distribution of different datasets, so there is no need to set various types of parameters, and no necessary parameters need to be set initially. The method is based on the MCMC theory, has good robustness, and is easy to use; 2) it can process cloud cover data of any size and is suitable for large-scale missing observations or even complete missing observations in a single layer and the interpolated image has no distortion and good spatial heterogeneity; and 3) it does not run longer due to the increase in missing observations. We innovatively use the early stopping mechanism on the Bayesian factorization method to reduce the unnecessary running time of the method and combined with the tensor restack to further reduce the running time.

Through experiments, we can determine the applicable scope of the HTR-BTTF method, small areas (space size less than 500×500), long-term sequences, and large areas affected by cloud cover. Future research should test the feasibility of the proposed method on other time-series remote sensing data and limit the maximum and minimum interpolation values, such as NDVI ranging from -1 to 1. In addition, the use of high performance computing (HPC) to solve the interpolation of large-area, long-term remote sensing data is worthy of further study.

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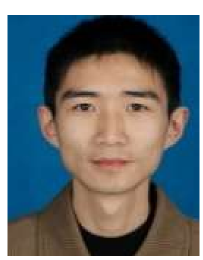

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